Assume/Guarantee Contract of `GetSSData`

June 24, 2026

1 Variable Declaration

Maps : `variables` in Code $\mapsto$ variables in Formal Specification

Alphabet :
royaw′, piyaw′, flgSp′, ADDR_AD_START, ADDR_AD_START′, AD_SS_LO4, AD_SS_LO4′, AD_SS_HI8, AD_SS_HI8′, SP_ADDR, SP_ADDR′, SP_BIT, SP_BIT′, chnl, chnl′, clock, clock′

Inputs : None

Outputs :

- `royaw` $\mapsto$ royaw : $\mathbb{R}$
- `piyaw` $\mapsto$ piyaw : $\mathbb{R}$
- `flgSp` $\mapsto$ flgSp : $\mathbb{N}$

In-Outputs : None

States : None

Parameters :

- `ADDR_AD_START` $\mapsto$ ADDR_AD_START : $\mathbb{N}$
- `AD_SS_LO4` $\mapsto$ AD_SS_LO4 : $\mathbb{N}$
- `AD_SS_HI8` $\mapsto$ AD_SS_HI8 : $\mathbb{N}$
- `SP_ADDR` $\mapsto$ SP_ADDR : $\mathbb{N}$
- `SP_BIT` $\mapsto$ SP_BIT : $\mathbb{N}$
- `chnl` $\mapsto$ chnl : $\mathbb{N} \rightharpoonup \mathbb{N}$

Clocks : clock, clock′ : $\mathbb{R}_{\geq 0}$

2 AG Contract

Assumptions :

$$Assumptions \triangleq \mathsf{len}(\mathsf{chnl}) = 2$$

Guarantees :

$$\begin{aligned}
Guarantees &\triangleq Check_1 \wedge Check_2 \\
Check_1 &\triangleq \mathsf{ADDR_AD_START}' = \mathsf{ADDR_AD_START} \\
&\quad \wedge \mathsf{AD_SS_LO4}' = \mathsf{AD_SS_LO4} \\
&\quad \wedge \mathsf{AD_SS_HI8}' = \mathsf{AD_SS_HI8} \\
&\quad \wedge \mathsf{SP_ADDR}' = \mathsf{SP_ADDR} \\
&\quad \wedge \mathsf{SP_BIT}' = \mathsf{SP_BIT} \\
&\quad \wedge \mathsf{chnl}' = \mathsf{chnl} \\
&\quad \wedge \mathsf{clock}' = \mathsf{clock} + 5.4 \times 10^{-6} \\
Check_2 &\triangleq \exists \mathsf{wd}_0, \mathsf{wd}_1 : \mathbb{N} \cdot \\
&\quad \begin{pmatrix} \mathsf{royaw}' = \mathsf{DiviationToFloat}(\mathsf{wd}_0, \texttt{0xFFF}, \texttt{0x800}, 0.00244140625) \\ \wedge\ \mathsf{piyaw}' = \mathsf{DiviationToFloat}(\mathsf{wd}_1, \texttt{0xFFF}, \texttt{0x800}, 0.00244140625) \end{pmatrix}
\end{aligned}$$